Literature Status: Fatou-Version Isometry Question for Symmetric Operator Spaces

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Status. **literature_already_answered**, with the scope caveat that the later theorem is stated for strongly symmetric operator spaces having the Fatou property.

Original source. Huang and Sukochev, *Hermitian operators and isometries on symmetric operator spaces*, arXiv:2101.03667; later JEMS 26 (2024), 3287–3325. The extracted target is the remark after Example 4.12, near source lines 2496–2502, which asks whether Theorem `th:iso` remains true under the weaker hypothesis that the symmetric operator spaces have Fatou norms only. The original theorem classifies surjective isometries under order-continuity assumptions.

Supporting answer. Fang, Guo, Huang, and Sukochev, *Isometric Structure in Noncommutative Symmetric Spaces*, arXiv:2512.20972. In the introduction, immediately before Theorem `11212`, the authors write that a goal of the paper is to treat the case where the norm on $E(\mathcal{M}, \tau)$ is not necessarily order continuous. Theorem `11212` states that if $\mathcal{M}_1, \mathcal{M}_2$ are atomless σ-finite von Neumann algebras, or σ-finite atomic von Neumann algebras whose atoms have the same trace, and $E(\mathcal{M}_1, \tau_1)$, $F(\mathcal{M}_2, \tau_2)$ are strongly symmetric operator spaces having the Fatou property and not proportional to L_2, then every surjective isometry has the same elementary Jordan form as in the Huang–Sukochev theorem, with $\sigma(F, F^\times)$-convergent sums in the semifinite case and a finite-trace simplification.

Identification. This is a separate later paper, not a same-paper answer. It cites the Huang–Sukochev JEMS paper as the order-continuous case, explicitly positions itself as handling the non-order-continuous case, and proves the corresponding elementary-isometry theorem under Fatou-property hypotheses. Therefore the 2101.03667 remark is no longer a new-proof target in the standard strong-symmetric/Fatou-property formulation.

Limitations. The later statement uses "Fatou property" and "strongly symmetric" language. This packet records a full literature answer for that formulation and does not separately settle any possible convention in which a bare "Fatou norm" assumption is strictly weaker than the hypotheses of Theorem `11212`.

Evidence searched. Local cheap indexes had no prior packet for arXiv:2101.03667 or arXiv:2512.20972. The local parsed source for arXiv:2512.20972 contains the explicit answer statement at source lines 391–412. Web search for the title and keywords located the same arXiv preprint and public metadata for the original JEMS article.

References

[1] J. Huang and F. Sukochev, *Hermitian operators and isometries on symmetric operator spaces*, arXiv:2101.03667; J. Eur. Math. Soc. 26 (2024), 3287–3325.

[2] K. Fang, T. Guo, J. Huang, and F. Sukochev, *Isometric Structure in Noncommutative Symmetric Spaces*, arXiv:2512.20972.